TEXORA · LECTURE SHEET · COURSE GUIDE

Industrial Visit: Report Writing and Presentation Guide

How to turn one factory visit into the full 100 marks of TE-2210: what to observe department by department, the data to collect while you are on the floor, and the structure of the report and the presentation that the marks are actually paid for.

TE-2210 · INDUSTRIAL VISIT · GUIDE SHEET · LEVEL 2 TERM 2

1. The course in brief

TE-2210 requires every student to **visit at least one textile industry** for practical experience of a working mill. There are no lectures and no machines of your own in this course — the factory is the classroom, and the evidence you bring back is the coursework. The marks distribution is fixed by the syllabus:

COMPONENT	WEIGHT	WHAT EARNS THE MARKS
Visit report	**60 %**	A complete, accurate, well-structured written report based on your own observations and collected data (Section 5)
Presentation	**40 %**	A clear, timed talk on the visit, defended in viva questioning (Sections 6–7)

Everything in this sheet follows from that split: the visit must be *worked*, not just attended. A student who walks the floor with a notebook and a question list (Sections 3–4) collects in three hours what the report and the viva will ask for over the following weeks.

2. Before and during the visit — working rules

- **Prepare the question list in advance** from Sections 3–4, customised to the type of mill (spinning, weaving, composite, garment). Review the relevant theory sheets (TE-2201, TE-2205, TE-2207) the night before so machine names and process terms are already familiar.
- **Carry** a notebook and pen (phones may be restricted on the floor; ask before photographing — many mills prohibit it, and a refused camera must never become an argument).

- **Safety and dress:** follow the mill's rules exactly — no loose clothing or open hair near rotating machinery, stay with the guide, do not touch running machines, settings or materials. You are a guest in a production environment.
- **Record numbers, not impressions.** "A big dyeing section" earns nothing; "14 jet dyeing machines, 8 working, liquor ratio 1:8" is report material. Where a figure is given verbally by the guide, note who gave it and mark it as reported, distinguishing it from what you saw yourself.
- **Note the flow.** At every department, ask the same three questions: what comes in, what is done to it, what goes out and where. The answers, department by department, are the skeleton of both the report and the presentation.

3. What to observe, department by department

Observe the departments the visited mill actually has; a composite mill may show all of these, a specialised mill only its own chain. For each department, record the machines (Section 4), the process sequence, and one quality point the department is responsible for.

DEPARTMENT	WHAT TO OBSERVE
Raw material store	Fibre/yarn types and counts stocked, bale/lot identification, storage conditions, how material is issued to production
Spinning (if present)	The machine line in order — blowroom, card, draw frame, lap former/comber, simplex, ring frame, winding; counts being spun; which machines are running and which idle
Weaving / knitting (if present)	Preparation first (winding, warping, sizing, drawing-in) then the looms/machines: type (shuttle, projectile, rapier, air-jet; or circular/flat knitting), fabric construction being produced, loom speed and efficiency displays
Dyeing and finishing — wet processing (if present)	Pretreatment line (singeing, desizing, scouring, bleaching, mercerising), dyeing machines (jet, jigger, pad), printing if present, finishing (stenter, sanforising, calendaring); water source and the effluent treatment plant (ETP) — its type and capacity
Garment / apparel (if present)	Marker and cutting section, sewing lines (line layout, operations sequence, machines per line), finishing, pressing, inspection and packing; the product being made and its buyer if disclosed
Quality control laboratory	Tests performed (fibre/yarn/fabric), instruments in use (e.g. evenness tester, strength tester, colour matching cabinet), standards followed, where in the flow testing happens

Utilities and maintenance	Boiler, generator/substation, compressor, humidification (spinning/weaving), the maintenance workshop and spare-parts store; power source and any captive generation
Warehouse and dispatch	Finished-goods storage, packing forms, labelling, how orders are assembled and dispatched

4. Data to collect — the checklist

DATA GROUP	ITEMS TO CAPTURE
Company facts	Name, location, year established, type of mill (spinning / weaving / composite / garment), approximate workforce, main products and markets
Machines	For each major machine: name, maker and country if on the plate, number installed and number running, key specification (e.g. ring frame spindles and spindle speed; loom width and rpm; dyeing machine capacity)
Materials and counts	Fibre/yarn counts in process (Ne, tex, denier as used), fabric constructions (ends/picks, GSM), dye classes used
Production	Daily/monthly production figures as reported (kg of yarn, metres of fabric, pieces of garment), shifts per day, machine efficiencies where displayed or reported
Quality points	The checks each department performs, the faults most commonly found, and what happens to rejected material
Utilities and environment	Water consumption source, ETP type and capacity, boiler fuel, power arrangement, waste disposal route
People and organisation	Departments and their heads as introduced, shift pattern, training or welfare facilities noticed (canteen, medical, safety signage)

If a figure cannot be obtained (commercially sensitive figures are often withheld), write that down too — "production figures not disclosed by the management" is itself an accurate observation, and inventing a number is the one unforgivable error in a visit report.

5. The report — structure

The report follows the material through the mill. Build the process-flow diagram first (Figure 1); every chapter then expands one box of it.

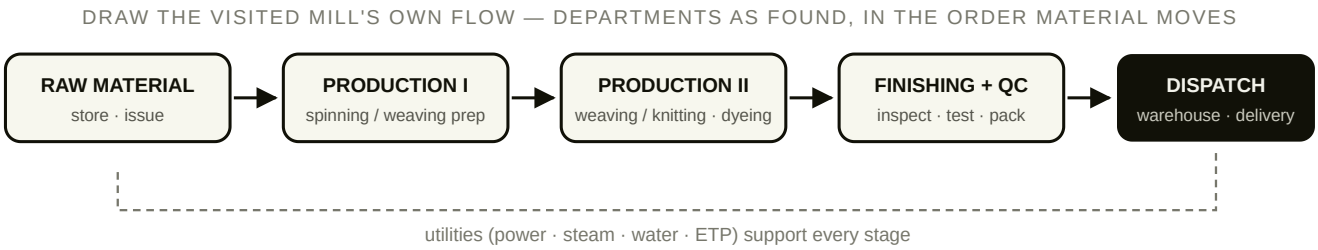


Figure 1 — The generic process flow of a textile mill. In your report, replace each box with the visited mill's actual departments and machine lines, in the order the material actually moves; departments the mill does not have are omitted, not invented.

1. **Title page** — report title, course code TE-2210, your name, roll/ID, department and institution, the industry visited, date of visit, date of submission.
2. **Acknowledgement** — brief thanks to the institute, the course teacher, and the mill's management and guide by name/role.
3. **Table of contents** — chapters with page numbers.
4. **Introduction** — purpose of industrial visits in textile engineering study; what this visit set out to see.
5. **Company profile** — the company facts of Section 4: identity, type, products, capacity, organisation in outline.
6. **Process flow** — the flow diagram (Figure 1 adapted) with a paragraph explaining the material's route through the mill.
7. **Department-wise observations** — the report's core: one section per department visited, following Section 3 — machines, process, counts/production, quality points, each with the collected data of Section 4.
8. **Findings** — what the visit demonstrated: links between theory and the floor (name the theory — e.g. "the comber settings of TE-2201 were seen at…"), strengths of the mill's practice, anything unexpected.
9. **Limitations** — honestly stated: departments not shown, data not disclosed, one visit's short duration.
10. **Conclusion** — what you can now do or understand that you could not before the visit; the visit's value to your remaining study.
11. **References** — the textbooks and course sheets used to interpret observations, and any company literature provided.

REPORT DISCIPLINE

Write the department sections within two days of the visit, while notebook entries still read clearly. Every number in the report must trace to a notebook entry — observed, read from a

> machine display, or marked "as reported by the guide". Photographs may be included only if the mill permitted them.

6. The presentation — structure and slide plan

The presentation compresses the report into a timed talk (confirm the allotted minutes with the course teacher; plan for about 8–10 minutes). One idea per slide; slides carry diagrams, tables and permitted photographs — the speaking carries the explanation.

SLIDE	CONTENT	SPEAKING POINT
1	Title — course, industry visited, your name/ID	Introduce yourself and the visit in two sentences
2	Company profile — type, products, scale	Place the mill in the industry: what it makes and for whom
3	Process flow diagram	Walk the audience along the material's route — this slide organises everything after it
4–6	One slide per major department — machines, one photo or table each	Per department: what comes in, what is done, what goes out, one number you collected
7	Production and quality data — your best table	The figures you verified, and what they show about the mill
8	Findings — theory seen in practice	Two or three explicit links to TE-2201 / TE-2205 / TE-2207 topics
9	Limitations and conclusion	What you could not see, and what the visit changed in your understanding
10	Thank-you / questions slide	Close, thank the mill and teachers, invite questions

- **Rehearse aloud, timed.** Most marks are lost by talks that are cut off mid-way; a rehearsed 9 minutes beats an unrehearsed 15.
- **Know your own numbers.** Every figure on a slide may be questioned in the viva — if you cannot explain a number's source, it should not be on the slide.
- **Prepare for the viva from Section 7** — the questions there are the ones this presentation reliably draws.

Glossary

Composite mill — a mill combining several production stages (e.g. spinning + weaving + dyeing) under one management. **ETP** — effluent treatment plant; treats the mill's waste water before discharge. **Liquor ratio** — the ratio of dye-bath water weight to fabric weight in wet processing. **GSM** — grams per square metre, the standard statement of fabric weight. **Captive power** — electricity the mill generates for itself. **Doff** — the removal of a full package (cop, bobbin, roll) from a machine and its replacement with an empty one.

Key points

• TE-2210 marks: Report 60 %, Presentation 40 % — the visit must be worked with a notebook and a question list, not merely attended.
• In every department ask: what comes in, what is done, what goes out — the answers form the report's skeleton.
• Record numbers with their sources; an honestly noted "not disclosed" is worth more than an invented figure.
• Build the report around the mill's own process-flow diagram; build the presentation around the same diagram, one idea per slide, rehearsed to time.
• The viva examines whether the report is truly yours — every number you present must be explainable.

7. Practice (presentation and viva preparation)

Prepare spoken answers of three to five sentences each; brief guidance follows.

1. Describe the visited mill's complete process flow, from raw material to dispatch, without looking at your slides.
2. Which single machine impressed you most, and what setting or specification of it can you quote?
3. The examiner points to any figure in your report: where did this number come from — observed, displayed, or reported?
4. State one place where mill practice differed from your textbook expectation, and explain the mill's reason.
5. What does the mill's ETP do, and why is it compulsory for a dyeing unit?
6. If you could revisit one department for a full day, which and why — what would you measure that a single visit could not show?

Answer key

1. Your answer must follow the material, department by department, in the mill's real order (Figure 1 adapted) — hesitation here signals a copied report; fluency is the strongest evidence the visit was

yours. Practise this as a 60-second spoken summary.

2. Choose a machine you genuinely studied (comber, loom, jet dyeing machine…) and quote one hard fact — a speed, a setting, a count — with its effect. A specific number defended beats a general admiration.
3. Answer with the source class and the detail: "read from the loom's efficiency display", "weighed/collected during the study", "given by the production manager". Never upgrade a reported figure into an observed one under pressure — examiners test exactly this honesty.
4. Good answers name a real difference — e.g. a mill running a coarser count than its machines' best, for order economics; or manual inspection where a textbook shows automatic — and give the mill's practical reason (cost, order mix, maintenance). The point is that practice optimises for the whole business, not for one textbook variable.
5. The ETP treats dyeing and finishing effluent — removing colour, chemicals and suspended solids and bringing pH and BOD/COD within discharge limits; it is compulsory because untreated wet-processing effluent is a major water pollutant and discharge without treatment breaches environmental law and buyer compliance codes.
6. Any department is acceptable if the reason is a measurement over time: efficiency across a full shift, end-breakage behaviour, dyeing shade variation between lots — phenomena a three-hour visit necessarily misses. This question tests whether you understand the limits of your own evidence.
